HOMOTOPIES OF PRODUCTS OF SIMPLICIAL OBJECTS

THE STACKS PROJECT AUTHORS

Editorial note. The counted statement is the author's own lemma environment labelled lemma-products-homotopy (source lines 4908-4925); the proof environment begins two lines below it and is its own proof, closing at source line 4949.

Editorial note. No \ref or \cite occurs anywhere inside the retained span, so the package declares neither a source pointer nor a bibliography entry; products of simplicial objects and homotopies are the author's surrounding development in the same chapter.

Editorial note. This card lies in the half of the chapter assigned to this lane under the line-number split (statement start >= 3601); the candidate was selected with that filter before reading, per the Lead's instruction.

Editorial note. The retained preamble is the chapter's own minimal preamble (sources/stacks-simplicial/preamble.tex, taken with its real line count), a verbatim excerpt of the native Xy-pic setup, the native theorem declarations with the mathematical macros, and the project's own no-op reference environments; no macro was invented or redefined.

Editorial scope. One original result of the Stacks Project's Simplicial Objects chapter, counted once: products of simplicial objects carry homotopies componentwise, and in the finite case conversely.

Allowed standard prerequisites (editorial).

- Simplicial object and its product. A simplicial object of a category is a contravariant functor on the simplex category, and its product over an index set is computed degreewise when it exists.. The product X is used degreewise throughout the proof (source lines 4910-4912, 4927-4940).
- Homotopy between morphisms of simplicial objects. Two morphisms of simplicial objects are homotopic when there is a family of maps in each degree satisfying the usual face and degeneracy identities expressing the homotopy.. The lemma is about producing and extracting such families (source lines 4914-4924, 4927-4948).
- Canonical projections and finite products. A map into a product is determined by its components via the canonical projections, and a finite product can be manipulated degree by degree.. Part (3) extracts the homotopies of the factors from a homotopy of the product maps using the projections (source lines 4920-4924, 4941-4948).

Example 0.1. Suppose in the situation above $a = b$. Then there is a *trivial* homotopy from a to b, namely the one with $h_{n,i} = a_n = b_n$.

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Remark 0.2. Let $\mathcal{C}$ be any category (no assumptions whatsoever). Let U and V be simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be morphisms of simplicial objects of $\mathcal{C}$. A *homotopy from a to b* is given by morphisms[1] $h_{n,i} : U_n \to V_n$, for $n \geq 0$, $i = 0, \ldots, n+1$ satisfying the relations of Lemma 0.4. As in Definition 0.3 we say the morphisms a and b are *homotopic* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}. Clearly, if $F : \mathcal{C} \to \mathcal{C}'$ is any functor and $\{h_{n,i}\}$ is a homotopy from a to b, then $\{F(h_{n,i})\}$ is a homotopy from $F(a)$ to $F(b)$. Similarly, if a and b are homotopic, then $F(a)$ and $F(b)$ are homotopic. Since the lemma says that the newer notion is the same as the old one in case finite coproduct exist, we deduce in particular that functors preserve the original notion whenever both categories have finite coproducts.

Definition 0.3. Let $\mathcal{C}$ be a category having finite coproducts. Suppose that U and V are two simplicial objects of $\mathcal{C}$. Let $a, b : U \to V$ be two morphisms.

(1) We say a morphism
$$h : U \times \Delta[1] \longrightarrow V$$
is a *homotopy from a to b* if $a = h \circ e_0$ and $b = h \circ e_1$.
(2) We say the morphisms a and b are *homotopic* or are *in the same homotopy class* if there exists a sequence of morphisms $a = a_0, a_1, \ldots, a_n = b$ from U to V such that for each $i = 1, \ldots, n$ there either exists a homotopy from a_{i-1} to a_i or there exists a homotopy from a_i to a_{i-1}.

Lemma 0.4. *In the situation above, we have the following relations:*

(1) *We have $h_{n,0} = b_n$ and $h_{n,n+1} = a_n$.*
(2) *We have $d^n_j \circ h_{n,i} = h_{n-1,i-1} \circ d^n_j$ for $i > j$.*
(3) *We have $d^n_j \circ h_{n,i} = h_{n-1,i} \circ d^n_j$ for $i \leq j$.*
(4) *We have $s^n_j \circ h_{n,i} = h_{n+1,i+1} \circ s^n_j$ for $i > j$.*
(5) *We have $s^n_j \circ h_{n,i} = h_{n+1,i} \circ s^n_j$ for $i \leq j$.*

Conversely, given a system of maps $h_{n,i}$ satisfying the properties listed above, then these define a morphism h which is a homotopy from a to b.

Proof. Omitted. You can prove the last statement using the fact, see Lemma 0.5 that to give a morphism of simplicial objects is the same as giving a sequence of morphisms h_n commuting with all d^n_j and s^n_j. □

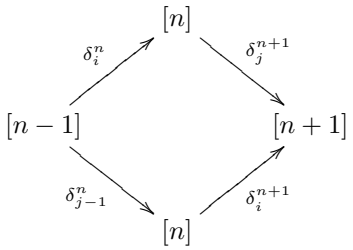
Lemma 0.5. *The category Δ is the universal category with objects $[n]$, $n \geq 0$ and morphisms δ^n_j and σ^n_j such that (a) every morphism is a composition of these morphisms, (b) the relations listed in Lemma 0.6 are satisfied, and (c) any relation among the morphisms is a consequence of those relations.*

Proof. Omitted. □

Lemma 0.6. *The morphisms δ^n_j and σ^n_j satisfy the following relations.*

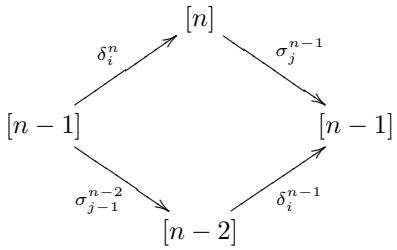
[1] In the literature, often the maps $h_{n+1,i} \circ s_i : U_n \to V_{n+1}$ are used instead of the maps $h_{n,i}$. Of course the relations these maps satisfy are different from the ones in Lemma 0.4.

(1) *If* $0 \leq i < j \leq n+1$, *then* $\delta_j^{n+1} \circ \delta_i^n = \delta_i^{n+1} \circ \delta_{j-1}^n$. *In other words the diagram*



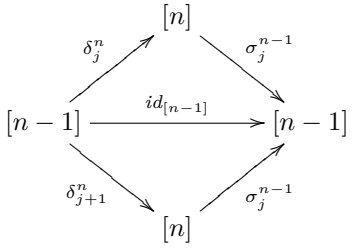
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(2) *If* $0 \leq i < j \leq n-1$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_i^{n-1} \circ \sigma_{j-1}^{n-2}$. *In other words the diagram*



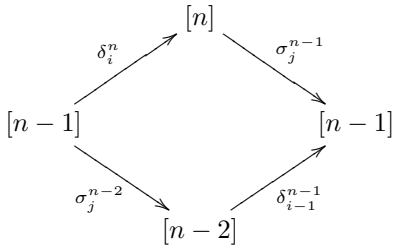
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(3) *If* $0 \leq j \leq n-1$, *then* $\sigma_j^{n-1} \circ \delta_j^n = id_{[n-1]}$ *and* $\sigma_j^{n-1} \circ \delta_{j+1}^n = id_{[n-1]}$. *In other words the diagram*



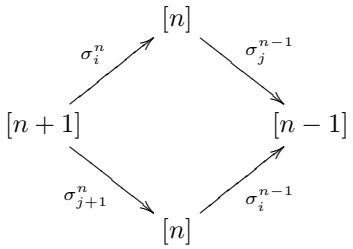
commutes.

(4) *If* $0 < j+1 < i \leq n$, *then* $\sigma_j^{n-1} \circ \delta_i^n = \delta_{i-1}^{n-1} \circ \sigma_j^{n-2}$. *In other words the diagram*



commutes.

(5) *If $0 \leq i \leq j \leq n-1$, then $\sigma_j^{n-1} \circ \sigma_i^n = \sigma_i^{n-1} \circ \sigma_{j+1}^n$. In other words the diagram*



commutes.

Proof. Omitted. □

Lemma 0.7. *Let $\mathcal{C}$ be a category. Let T be a set. For $t \in T$ let X_t, Y_t be simplicial objects of $\mathcal{C}$. Assume $X = \prod_{t \in T} X_t$ and $Y = \prod_{t \in T} Y_t$ exist.*

(1) *If X_t and Y_t are homotopy equivalent for all $t \in T$ and T is finite, then X and Y are homotopy equivalent.*

For $t \in T$ let $a_t, b_t : X_t \to Y_t$ be morphisms. Set $a = \prod a_t : X \to Y$ and $b = \prod b_t : X \to Y$.

(2) *If there exists a homotopy from a_t to b_t for all $t \in T$, then there exists a homotopy from a to b.*

(3) *If T is finite and $a_t, b_t : X_t \to Y_t$ for $t \in T$ are homotopic, then a and b are homotopic.*

Proof. If $h_t = (h_{t,n,i})$ is a homotopy from a_t to b_t (see Remark 0.2), then $h = (\prod_t h_{t,n,i})$ is a homotopy from $\prod a_t$ to $\prod b_t$. This proves (2).

Proof of (3). Choose $t \in T$. There exists an integer $n \geq 0$ and a chain $a_t = a_{t,0}, a_{t,1}, \ldots, a_{t,n} = b_t$ such that for every $1 \leq i \leq n$ either there is a homotopy from $a_{t,i-1}$ to $a_{t,i}$ or there is a homotopy from $a_{t,i}$ to $a_{t,i-1}$. If $n = 0$, then we pick another t. (We're done if $a_t = b_t$ for all $t \in T$.) So assume $n > 0$. By Example 0.1 there are is a homotopy from $b_{t'}$ to $b_{t'}$ for all $t' \in T \setminus \{t\}$. Thus by (2) there is a homotopy from $a_{t,n-1} \times \prod_{t'} b_{t'}$ to b or there is a homotopy from b to $a_{t,n-1} \times \prod_{t'} b_{t'}$. In this way we can decrease n by 1. This proves (3).

Part (1) follows from part (3) and the definitions. □